

Indian Institute of Engineering Science and Technology, Shibpur
BTech 4th Semester Mid-Semester Examination, February 2026

Computer Architecture and Organization-I (CS2202)

Full Marks: 30

Time: 2 Hours

Answer any three questions

1. Answer the following in the context of fixed point and floating point representation of real number.
 - (a) Disadvantage of fixed point representation of a real number.
 - (b) Smallest and largest positive number in the 32-bit format of fixed point representation.
 - (c) Normalized floating point number with example.
 - (d) Smallest and largest positive normalized floating point number in 32-bit format.
[2+2+2+(2+2)]
2. (a) What are the advantages and disadvantages of hardwired control unit over microprogrammed control unit.
 - (b) Define the following terms in the context of microprogrammed control unit.
(i) Microinstruction (ii) Microprogram (iii) Microoperation (iv) Control memory
 - (c) Compare horizontal and vertical format of microinstruction.
 - (d) Total number of microoperation is 46 and degree of parallelism is 2. What is the size of two subfields in 9-bit control field of microinstruction?
[2+4+2+2]
3. (a) Define absolute and relative addressing.
 - (b) What are the advantages and disadvantages of relative addressing over absolute addressing?
 - (c) Discuss about the difference between the following four instructions.
(i) ADD (ii) ADD T (iii) ADD X, T (iv) ADD X, X, T
 - (d) Discuss about the basic principle of Booth's multiplication algorithm.
[2+2+4+2]
4. (a) In the context of Von Neumann machine
 - (i) What is the size of instruction and format of instruction?
 - (ii) What is the size of data and format of data?
 - (iii) What is the function of the following high speed registers?
AC, MQ, DR, IBR, PC, IR
 - (b) Define instruction cycle, fetch cycle and execution cycle.
[2+2+3+3]